

UNDERSTANDING EMBEDDINGS IN NLP



How AI learns meaning from
text and data

WHAT ARE EMBEDDINGS?

Definition: Numerical representations of text in a multi-dimensional space.

Why it matters: Enables AI to understand relationships, context, and meaning.

Example:

- "King" - "Man" + "Woman" $\approx$ "Queen"
- Words with similar meanings are closer in vector space.

TYPES OF EMBEDDINGS

- ◆ **Word Embeddings**
- ◆ **Contextual Embeddings**
- ◆ **Sentence & Document Embeddings**
- ◆ **Multimodal Embeddings**

WORD EMBEDDINGS

What It Is

Word embeddings convert words into fixed-length numerical vectors based on their semantic similarity.

Models

- Word2Vec (CBOW & Skip-gram)
- GloVe (Global Vectors for Word Representation)
- FastText (Handles subword information)

Use Cases

- ✓ Sentiment analysis
- ✓ Named Entity Recognition (NER)
- ✓ Text classification

WORD EMBEDDINGS

Pros:

- ✓ Efficient & Lightweight – Pre-trained word embeddings are fast and require minimal computational power.
- ✓ Captures Semantic Relationships – Words with similar meanings are mapped closer in vector space.
- ✓ Widely Used & Pre-Trained Models Available – Saves training time and effort.

Cons:

- ✗ Context Ignorance – Treats words in isolation without considering the sentence context.
- ✗ Fails with Polysemy – The same word (e.g., bank in "river bank" vs. "financial bank") has a single representation.
- ✗ Fixed Vocabulary – Cannot handle out-of-vocabulary (OOV) words effectively.

CONTEXTUAL EMBEDDINGS

What It Is

Unlike word embeddings, contextual embeddings change word meanings based on context within a sentence.

Models

- BERT (Bidirectional Encoder Representations from Transformers)
- GPT (Generative Pre-trained Transformer)
- RoBERTa (A more robust version of BERT)

Use Cases

- ✓ Chatbots & virtual assistants
- ✓ Question answering systems
- ✓ Machine translation

CONTEXTUAL EMBEDDINGS

Pros:

- ✓ Context-Aware – Adjusts word meanings based on surrounding words.
- ✓ Handles Polysemy – Different vectors for the same word in different contexts.
- ✓ Improves NLP Accuracy – Great for chatbots, question-answering, and translation.

Cons:

- ✗ Computationally Expensive – Requires GPUs or TPUs for training and inference.
- ✗ Requires Large Datasets – Pre-training models need extensive corpora.
- ✗ Slow Inference Time – Especially for real-time applications.

SENTENCE & DOCUMENT EMBEDDINGS

What It Is

These embeddings represent entire sentences or documents as a single vector, preserving contextual meaning.

Models

- Universal Sentence Encoder (USE)
- Sentence-BERT (SBERT)
- Doc2Vec

Use Cases

- ✓ Semantic search (e.g., Google Search)
- ✓ Text summarization
- ✓ Document similarity and clustering

SENTENCE & DOCUMENT EMBEDDINGS

Pros:

- ✓ Captures Sentence-Level Meaning – Instead of focusing only on words.
- ✓ Ideal for Similarity Tasks – Helps in semantic search and clustering.
- ✓ Better Performance on Long Texts – Useful for paragraph- or document-level analysis.

Cons:

- ✗ Less Word-Level Precision – Doesn't provide detailed word relationships.
- ✗ Computationally Intensive – More complex than word embeddings.
- ✗ Context Dependence Varies – Some models still struggle with ambiguity.

MULTIMODAL EMBEDDINGS

What It Is

Multimodal embeddings combine text, images, and audio into a shared embedding space.

Models

- CLIP (Contrastive Language-Image Pretraining)
- VisualBERT
- LXMERT (for vision and language tasks)

Use Cases

- ✓ Image captioning
- ✓ Visual question answering
- ✓ Cross-modal search (e.g., searching an image using text)

MULTIMODAL EMBEDDINGS

Pros:

- ✓ Integrates Different Data Types – Text, images, audio, and more.
- ✓ Great for Cross-Domain Tasks – E.g., AI-powered image captioning, visual search.
- ✓ Enables Advanced AI Applications – Powers AR, VR, and real-time translation.

Cons:

- ✗ High Computational Demand – Requires more resources than text-based embeddings.
- ✗ Difficult to Train – Needs large and diverse datasets.
- ✗ Limited Generalization – May not perform well outside trained domains.

EMBEDDINGS

AI APPLICATIONS

✓ Semantic Search

Google, chatbots

✓ Recommendation Systems

Netflix, YouTube

✓ Text Classification

Spam detection, sentiment analysis

✓ Clustering

Topic modeling, news categorization



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